DATABASE MANAGEMENT 475/575 · STAGE 4

Almanac

A farm record book that knows what your crops are worth

Team Lesotho · Dylan Montgomery & Andrew Styles · October 28, 2026

THE IDEA

Three kinds of data, one shared key

Farm record book

What a grower planted, harvested, sold or lost, on which plot and when.

Auction prices

What each crop actually sold for at Ohio produce auctions, sale day by sale day, 2019 to 2026.

Plant taxonomy

A tree that identifies every crop precisely: order, family, genus, species, variety.

Every event and every price resolves to a **taxon**, so a grower’s history and the market’s history can be queried together.

REAL DATA

Collected, cleaned and loaded

33,667	520	2,993	107
auction prices	sale days, 2019–2026	taxonomy nodes	real Ohio farms

Prices: USDA AMS Market News reports for the Mount Hope Auction (391 sale days, 2019–2026) and County Line Produce Auction market reports (219 sale days, 2024–2026).

Taxonomy: GBIF Backbone Taxonomy. **Farms:** OpenStreetMap, with counties from the US Census.

Quality: 26 rows failed a CHECK constraint (average outside its own low–high range) and sit in a quarantine table with the reason. 85.8% of prices already match a product. 237 unit spellings map to 35 standard units; pecks and bushels are never averaged together.

DATABASE

12 tables, 15 foreign keys, 2 views

Reference data · 8 tables

observer, region, taxon, product, product_alias, market_lot, market_lot_quarantine, event_kind

taxon and **region** point to themselves: two hierarchies.

Grower data · 4 tables

farm, plot, crop_catalog, farm_event

Rules live in the schema: deleting a plot keeps its events; a taxon in use can’t be deleted; a quantity needs a unit.

PostgreSQL 16 · Python 3 · Flask · psycopg 3 · hand-written, parameterized SQL (no ORM) · Chart.js

LIVE DEMO

The basic functions, in the app

1 · Real data

Home page: row counts for every table and match rate per source.

2 · Insert

Add a crop, a plot, and a harvest event to a real farm.

3 · Update and delete

Edit the harvest; delete the plot and watch its event survive.

4 · Search

Crop search, the price curve, and harvest vs. market price.

SEARCH

Four queries, four questions

Page	Question it answers
Crop search	Which taxa or products match this text? Ranked by how much price evidence each has.
Price curve	What is this crop’s floor, mean and ceiling price in each month, across all seasons?
Activity history	What did this farm do in this date range, for this event kind and crop?
Harvest vs. market	**For each harvest, what did that crop sell for at auction within 30 days?**

The last one is our multi-table join: farm_event, farm, event_kind, taxon, product, product_alias and market_lot.

NEXT: ADVANCED FUNCTIONS

Three things no farm app can do

1 · RECURSIVE CTE

Taxonomic price roll-up

Pick any node, such as a family or order, and price everything beneath it, at any depth, in one query.

2 · WINDOW FUNCTION

Crop decision engine

Rank crops for a planting window across all years by price, discounted for volatility.

3 · DATA RESOLUTION

Label resolution ladder

Fetch and parse new reports automatically, then match the 1,445 unmatched labels in four rungs; stop on ambiguity, never guess.

PLAN

From here to the final demo

Week of	Work
Oct 26	Advanced Function 1 page; source comparison page
Nov 2	Advanced Functions 2 and 3, with the review screen
Nov 9	**Stage 5 report; site goes public and we start inviting growers**
Nov 16	Collect real user records; fix bugs from feedback
Nov 23	Feature freeze, full testing, rehearsal
Nov 30	Final demo: basic functions and all three advanced functions

Questions?

Almanac · Team Lesotho

github.com/dylcmonty/CPSC475